Online Resource 1: Supplementary Material

Field-cooperative neural topic modelling of civil engineering literature: missing-abstract inference

Applied Intelligence

Auxiliary Experiments, Controls and Metric Definitions

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1 Auxiliary experiments and controls

1.1 Long-document truncation control

The GitHub background field has a long tail (up to 20532 tokens after preprocessing) while the sentence-embedding baselines truncate at 128 tokens. To check whether the ordering of the main tables is caused by this truncation, the three baselines were re-run with window pooling: each document is split into 128-token windows with the model's own tokenizer, the windows are encoded separately and their vectors averaged, so the embeddings cover the full document; all other settings are unchanged. Table S1 reports the result.

Window pooling raises BERTopic from 0.540 to 0.582 and CombinedTM from 0.643 to 0.656 on GitHub, and BERTopic from 0.488 to 0.518 while CombinedTM moves from 0.526 to 0.488 on AG News. Even at their best (GitHub CombinedTM 0.656, AG News BERTopic 0.518) the pooled baselines remain below the class TF–IDF read-out of FC-NTM (0.727 and 0.612), so the ordering of bag-of-words models and sentence-embedding baselines is not an artefact of truncation.

1.2 Within-corpus length-ratio ladder

The cross-corpus comparison mixes the length ratio with domain and genre. To separate part of this confound, the civil corpus is truncated to the first 60 and 30 words of every abstract (length ratio $11.29 \to 3.99 \to 2.02$) and the full, no-cross and no-mask configurations are re-run for the same 6000/2000 record IDs and the same 7685-word vocabulary (700-epoch budget, all other settings unchanged). Table S2 reports the result; three observations follow. First, the β read-out of the full model decreases monotonically with truncation ($0.633 \to 0.595 \to 0.534$) while the class TF–IDF read-out does not ($0.636 \to 0.645 \to 0.606$): the effect of truncation depends on the read-out. Second, the paired difference of removing cross-field reconstruction does not change monotonically with the length ratio (β: $-0.022 \to +0.006 \to -0.006$; class TF–IDF: $-0.019 \to +0.001 \to -0.023$). Third, the cost of removing word masking also fails to follow the ratio (β: $-0.010 \to -0.002 \to -0.027$; class TF–IDF: $-0.011 \to -0.017 \to -0.025$), and the read-out with the smallest cost differs. Under a controlled change of the abstract length alone, component importance is therefore neither monotone in the ratio nor consistent across read-outs, which agrees with the cross-corpus result that the length ratio is a description rather than a pre-training criterion.

Table S1: Long-document control for the sentence-embedding baselines (C_V, seeds 11/23/37, mean±standard deviation). "Truncated" encodes only the first 128 tokens (the setting used in the main tables); "window pooling" splits a document into 128-token windows with the model's own tokenizer and averages the window vectors, so every token enters exactly one window. Top2Vec uses Doc2Vec document vectors and does not accept window pooling.

model	GitHub Truncated	GitHub Window pooling	AG News Truncated	AG News Window pooling
BERTopic	0.540±0.015	0.582±0.009	0.488±0.022	0.518±0.011
Top2Vec	0.493±0.009	n/a	0.498±0.032	n/a
CombinedTM	0.643±0.022	0.656±0.018	0.526±0.037	0.488±0.030

Table S2: Within-corpus length-ratio ladder on the civil corpus ($K = 30$, seeds 11/23/37). ΔC_V is the paired difference of removing one component relative to the full model (negative means the component matters); parentheses count positive seeds. The length ratio is the mean token ratio of abstracts to titles over the same records and vocabulary.

Variant(Length ratio)	read-out	Full C_V	− Cross-field recon. ΔC_V	− Word masking ΔC_V
Original(11.29)	β	0.633	−0.022 (0/3)	−0.010 (1/3)
Original(11.29)	Class TF–IDF	0.636	−0.019 (0/3)	−0.011 (0/3)
abs60(3.99)	β	0.595	+0.006 (3/3)	−0.002 (2/3)
abs60(3.99)	Class TF–IDF	0.645	+0.001 (1/3)	−0.017 (0/3)
abs30(2.02)	β	0.534	−0.006 (1/3)	−0.027 (0/3)
abs30(2.02)	Class TF–IDF	0.606	−0.023 (0/3)	−0.025 (0/3)

1.3 Controls for the occlusion protocol

Three controls test what the relative degradation can show: (i) a constant unigram model that only uses training term frequencies and reads no test document; (ii) a shuffled summary, where the summary of each document is replaced by that of another document while the model is unchanged; and (iii) the same architecture trained without whole-field dropout. All three use the same masks, seeds and per-token NLL. Tables S3 and S4 report the results.

Three conclusions follow. First, the constant unigram model has a relative degradation of almost one on all three corpora (0.999–1.001) and its absolute NLL at full occlusion is lower than that of FC-NTM on GitHub and the civil corpus (7.382 versus 7.961 and 7.875 versus 8.160) and close on AG News (7.809 versus 7.828): a model that reads no test input can be "stable" under this ratio, and the cross-model comparison of absolute likelihood mixes in the calibration of the word distributions. The ratio is therefore a diagnostic under a specified baseline set, not evidence of field complementarity by itself. Second, replacing the summary by that of another document raises the relative degradation from 1.029/1.018/1.010 to 1.091/1.096/1.084 and worsens the absolute NLL (8.567/8.622/8.997 versus 7.961/7.828/8.160): the model really uses the *content* of the summary field, which supports "the summary participates in inference" more directly than the ratio itself. Third, removing training-time dropout leaves the ratio almost unchanged (1.028/1.020/1.012 versus 1.029/1.018/1.010). A same-budget single-encoder control is not compared under this occlusion protocol, so the behaviour on these curves cannot be attributed to the two-stream structure alone.

1.4 Sensitivity of the β read-out

The main tables report the generative β read-out, $\mathrm{softmax}(\mathrm{BN}(B_k))$ for a unit topic input; the raw decoder rows B_k are reported as a parameter-level view. The document generative distribution of FC-NTM is $\mathrm{softmax}(\mathrm{BN}(\theta B))$, and the two readings are not equivalent. Feeding a unit topic input

Table S3: Mechanism controls for the partial-occlusion protocol (seeds 11/23/37). Each cell shows the mean±standard deviation of the relative degradation NLL(100%)/NLL(25%) and the absolute NLL at full occlusion. The constant unigram model reads no test input; "shuffled summary" replaces each summary by that of another document; "true summary" is the main model.

corpus	constant unigram	shuffled summary(FC-NTM)	FC-NTM(true summary)
GitHub	0.999±0.001 / 7.382	1.091±0.002 / 8.567	1.029±0.003 / 7.961
AG News	1.001±0.000 / 7.809	1.096±0.001 / 8.622	1.018±0.001 / 7.828
civil corpus	1.000±0.001 / 7.875	1.084±0.001 / 8.997	1.010±0.001 / 8.160

Table S4: Effect of training-time field dropout on the partial-occlusion curve (relative degradation, seeds 11/23/37). The two columns are the same architecture with and without whole-field dropout during training.

corpus	dropout(mainmodel)	No dropout
GitHub	1.029±0.003	1.028±0.007
AG News	1.018±0.001	1.020±0.003
civil corpus	1.010±0.001	1.012±0.001

e_k into the decoder gives the generative distribution, whose top-10 words overlap the raw read-out with an average Jaccard of only 0.36–0.42 and whose first word differs for 56%–74% of the topics; the C_V and NPMI of the same models also drop from the raw values (GitHub 0.701→0.625 and 0.233→0.192; AG News 0.586→0.541 and 0.137→0.100; civil corpus 0.633→0.584 and 0.154→0.118), as Table S5 shows. A translation test makes the same point: adding one shared per-word vector to all topic rows and shifting the BatchNorm running mean leaves the generative distribution pointwise unchanged (maximum probability difference $\leq 5 \times 10^{-7}$) while 100% of the raw top-10 lists change. The class TF–IDF read-out is built from θ and term counts and is not affected; both β readings are reported, and the generative reading is used in the main tables because it reflects the model's generative semantics.

2 Data and metric notes

Training details of the baselines

The four baseline families use the following settings. *Gibbs sampling*: BTM is a collapsed Gibbs sampler over biterms (word pairs within a window of 12 tokens, at most 150 sampled pairs per document, $\alpha = 50/K$, $\beta = 0.01$, 300 full sweeps); LDA is fitted by gensim's batch variational Bayes (50 passes, 120 iterations per pass, symmetric α). *Matrix factorisation*: NMF minimises the Kullback–Leibler divergence between the count matrix and WH with multiplicative updates (600 iterations, tolerance 10^{-4}). *Neural variational models*: ProdLDA, ETM, CTM-lite and FC-NTM share the Adam loop, mini-batch size and budget (700 epochs, 20% held-out early stopping); the baselines use learning rate 2×10^{-3} and the main FC-NTM configuration 5×10^{-4} selected on development data, and the official CombinedTM keeps its own loop (Adam, learning rate 2×10^{-3}, batch size 64, 700 epochs, early stopping on its own held-out split). *Embedding clustering*: BERTopic runs UMAP followed by HDBSCAN with fixed random seeds, and Top2Vec fits Doc2Vec document vectors followed by HDBSCAN; both choose their own topic numbers and are reported for reference only.

Table S5: Sensitivity of the β read-out (main checkpoints of the three corpora, means over seeds 11/23/37). "raw" reads the decoder parameter rows B_k; "BN" reads the generative distribution softmax(BN(B_k)) for a unit topic input; Jaccard is the average per-topic overlap of the top-10 words and the top-1 change rate the share of topics whose first word differs.

corpus	raw C_V	BN C_V	raw NPMI	BN NPMI	top-10 Jaccard	top-1 change rate
GitHub	0.701	0.625	0.233	0.192	0.369	0.74
AG News	0.586	0.541	0.137	0.100	0.360	0.62
civil corpus	0.633	0.584	0.154	0.118	0.421	0.56

2.1 Data notes

After stop-word removal and vocabulary filtering, a small number of identical token sequences occur in both the training and the test split (GitHub 10 records, AG News 8 records, i.e. 0.50% and 0.11% of the test sets); they come from different original records and are a degeneration of the filtering rather than duplicated raw records, and the proportion is too small to change the conclusions. The GitHub background field has a long tail (8501 tokens in the training set and 20532 in the test set after preprocessing) that the sentence-embedding baselines truncate. A few records have an empty summary after preprocessing (GitHub 6 training and 1 test record; AG News 69 training and 98 test records, up to 1.29% of the test set); these records are kept in the main tables with an empty summary input, for which the model has no explicit fallback branch – the summary encoder still runs and produces a training-parameter-dependent posterior before gating with the background posterior (for the seed-11 checkpoint the maximum difference from a background-only posterior is 0.115 for the GitHub record and 0.261 for the AG News records, with the most probable topic unchanged in all but three cases). The dropout training and the occlusion experiments therefore validate *background* absence and should not be extrapolated to missing summaries. Design iteration used 400 test records (a 1000/400 subsample per corpus); the main text reports both the full test split and a development-free subset (1600 GitHub and 7200 AG News records) obtained by exact record-level matching, on which the coherence is recomputed for every stored model. That re-scoring removes direct overlap only and cannot undo the model and configuration choices already made, so it is a sensitivity analysis rather than an untouched confirmation set; an independent confirmation would require repositories or corpora not involved in the development.

2.2 Supplementary definitions of the metrics

$\mathrm{EXCL}_{\mathrm{KADJ}}$ is a K-corrected variant of the Bischof–Airoldi exclusivity: the usual EXCL is the mean column-normalised mass of the top ten words of each topic, and it is mapped to $\mathrm{EXCL}_{\mathrm{KADJ}} = (\mathrm{EXCL} - 1/K)/(1 - 1/K)$ so that random topics score about zero and fully exclusive topics about one. This is our definition rather than the original measure, and it does not remove the comparability issues between different K or between different score matrices, so the table only compares models with the same $K = 30$. For the FC-NTM-cTFIDF rows the value is computed on the class TF–IDF score matrix after row normalisation, which is not a probability distribution, so those rows are only ranked within the same read-out and are not compared with β rows. IRBO uses the finite-length extrapolated form, $\mathrm{RBO}_{\mathrm{ext}} = (1-p)\sum_{d=1}^{L} p^{d-1} A_d + p^L A_L$ with $p = 0.9$, common depth L and set overlap A_d at depth d, and topic diversity is $\mathrm{IRBO} = 1 - \overline{\mathrm{RBO}}_{\mathrm{ext}}$. Without the tail term even two identical ten-word topic lists would obtain a "similarity" of $1 - p^L = 0.651$ (diversity 0.349); the bias varies with the overlap at the final depth (it equals p^L for identical and zero for disjoint lists), so it is not a constant offset shared by all models.

Table S6: Example topics on GitHub issues (FC-NTM, seed 11, $K = 30$, best epoch 547 – the same model as in the main table and figure). The words are generated from the topic–word matrix: the class TF–IDF read-out is computed first and then truncated to the summary and background vocabularies, so no word lies outside the vocabulary; topic names are assigned manually.

topics(manual)	Summary-side words(top 10)	Background-side words(top 10)
Type systems and language services	type typescript function string types regression error property assignable array	type typescript function string types regression error property assignable array
DuckDB build and distribution	duckdb all python data sets distribution architecture linux required nightly	duckdb all python data reproducing sets distribution architecture linux required
Redis replication and clustering	master cluster redis node sentinel slave nodes docker failover command	master cluster redis node sentinel slave nodes docker failover command
Configuration and command semantics	redis command key set value some keys error one cluster	redis command key set value some keys error one cluster
Language feature proposals	javascript type typescript suggestion types existing runtime change feature syntax	javascript type typescript suggestion types wouldn existing runtime change feature
Event loop and cache metrics	idle addr sub flags cmd age multi free events name	qbuf psub oll obl idle addr sub flags cmd omem

3 Additional tables and figures

The table below lists the example topics referenced in the main text.

3.1 Civil topic case studies

The table below lists four readable topics of the seed-11 civil model with one representative record each, plus two failure cases found by scanning the labelled test records. In the first failure case a seismicity paper is routed by its title alone to the BIM topic, and the abstract restores the correct label; in the second, an entry whose title mentions corrosion is routed by both settings to the related "concrete reinforcement materials" topic instead of the corrosion topic. Routing uses the same majority-vote mapping protocol as the main archival task. The representative records of the four topics are selected by topic mass on the training split and are not held-out evidence; the two failure cases are test records. The mixed laboratory-materials topic is an interpretability problem (mixed vocabularies), not a routing failure — its representative record is routed correctly.

4 Additional tables for converted results

Table S7: Case studies on the civil corpus (FC-NTM, seed 11, β read-out; "title-only" and "full-field" give the routing decision of the same mapping protocol, and "miss" marks a decision different from the OpenAlex label; OpenAlex Work IDs are given in parentheses).

Topic(manual)	Top words(β read-out)	Representative record	OpenAlex label	title-only	full-field
Asphalt mixture design and performance	asphalt, mixture, binder, mixtures, temperature, test, performance, modified, stability, low	Synergetic Effect of NanoZnO and Trinidad Lake Asphalt for Antiaging Properties of SBS Modified Asphalt (W3007633598)	Asphalt Pavement Performance Evaluation	correct	correct
Seismic response of tunnels and piles	response, ground, motion, seismic, tunnel, earthquake, acceleration, foundation, pile, wave	Earthquake ground-motion parameters at stiff sites in Addis Ababa, Ethiopia (W2033407901)	Seismic Performance and Analysis	correct	correct
BIM and construction management	construction, information, bim, building, project, industry, management, projects, implementation, process	Exploring the Adoption of Building Information Modeling (BIM) in the Jordanian Construction Industry (W2624290857)	BIM and Construction Integration	correct	correct
Structural mechanics: shear, buckling, composite beams	load, capacity, finite, element, shear, beam, plate, composite, behavior, buckling	Out-of-Plane Secondary Bifurcation Buckling Behavior of Elastic Circle Pipe Arch (W2063862378)	Structural Load Bearing Analysis	correct	correct
Mixed laboratory materials topic (failure case)	diamond, wood, metal, cell, density, wear, segments, adsorption, matrix, saw	Pseudotsuga menziesii (Mirb.) Franco wood basic density in the Patagonia Argentina (W1511725581)	Wood Treatment and Properties	correct	correct
Failure case: title only misrouted	construction, information, bim, building, project, industry, management, projects, implementation, process	United States building code approach to variations in regional seismicity (W2525447080)	Seismic Performance and Analysis	miss	correct
Failure case: title only misrouted	strength, fiber, steel, concrete, mechanical, reinforced, fibers, specimens, compressive, tensile	Corrosion Behavior of Reinforcing steel in Concrete (W4387417777)	Concrete Corrosion and Durability	miss	miss

Table S8: Concatenated-field single-encoder control (seeds 11/23/37; same budget, vocabulary and read-outs). β rows use the generative softmax(BN($e_k W$)) read-out; Δ is FC-NTM minus the control, with positive pairs in parentheses.

Corpus	Read-out	FC-NTM	Concatenated	Δ (positive pairs)
GitHub	β	0.625±0.026	0.587±0.017	+0.037 (2/3)
GitHub	Class TF–IDF	0.727±0.022	0.710±0.020	+0.017 (1/3)
AG News	β	0.541±0.041	0.533±0.011	+0.008 (2/3)
AG News	Class TF–IDF	0.612±0.014	0.584±0.009	+0.028 (3/3)
Civil corpus	β	0.584±0.029	0.564±0.005	+0.020 (2/3)
Civil corpus	Class TF–IDF	0.636±0.014	0.606±0.014	+0.029 (2/3)

Table S9: Title-only routing to the 15 most frequent OpenAlex topics of the civil corpus (seeds 11/23/37; 857 test records, coverage 42.9%). The supervised TF–IDF classifier is a reference point on the same label set.

Method	Input	Accuracy	Macro-F1
TF–IDF logistic regression (supervised)	title	0.810	0.805
FC-NTM	title + abstract	0.719±0.016	0.676±0.032
FC-NTM	title only	0.664±0.004	0.624±0.013
Concatenated control	title only	0.653±0.012	0.614±0.034
BTM	title only	0.600±0.013	0.529±0.043
LDA	title only	0.532±0.039	0.458±0.057